

Assimilative 2-D Lagrangian Transport Model for the Estimation of Oil Leakage Parameters From SAR Images: Application to the Montara Oil Spill

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Abstract—Indonesia is particularly vulnerable to oil spill since Indonesian waters, which account for 80% of its territory, involve a very active maritime traffic and a large number of onshore and offshore oil platforms. Within the framework of Infrastructure Development Space Oceanography project, we address the operational synthetic aperture radar (SAR) based monitoring of oil spill in Indonesian waters from space. This work focuses on the assimilation of SAR observations into a 2-D Lagrangian trajectory model. The simulated surface oil trajectories are compared to the satellite observations in subsequent forecast cycles for veracity testing by estimating best leakage parameters of detected oil spills. As a case study, we consider a large oil spill event that took place in Indonesian waters in Fall 2009, referred to as the Montara oil spill. A novel methodology has been developed, which combines SAR-based oil spill detection using a Lagrangian analysis and numerical tools. Our model relies on four key parameters: wind-related and current-related drift parameters, and the origin and the duration of the oil leakage. Given an SAR-derived oil spill detection, a numerical inversion is used to optimize these model parameters, so that the simulated drift matches the SAR-derived observation. The demonstration of the relevance of the proposed scheme for the Montara oil spill and further discussion on operational interest for satellite-based oil spill monitoring can be useful in rapid response system in Indonesia.

Index Terms—Pollution, synthetic aperture radar (SAR).

I. INTRODUCTION

OIL spill is one of the most serious threats for all marine and coastal environments. Oil spill can cause high economical and ecological damages in marine and coastal ecosystems. Indonesia, as an archipelago country, is particularly vulnerable to oil spill since the national waters, which account for 80% of its territory, involve a very active maritime traffic and a large number of onshore and offshore oil platforms. The accidental or intentional releases of petroleum products are the two sources

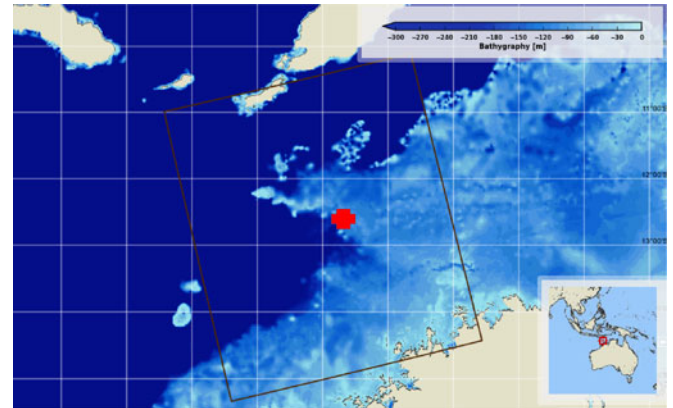


Fig. 1. Position of the Montara oil platform on the map of bathymetry; the red sign is the Montara wellhead position and the black box is our boundary box for oil trajectory simulation.

of oil pollution into the marine environment. Even though the pollution from voluntary rejection of waste oil and residual fuel are of high importance, for instance the equivalent of 50 Erika or 15 Prestige per year in the Mediterranean Sea [1], major oil spill accidental disasters such as Deepwater Horizon or Torrey Canyon oil spills still have devastating long-term consequences for the society, economically, environmentally, and socially [2].

The Montara oil spill in 2009 stressed the urgent needs for the operational monitoring of oil spill in Indonesian waters. The operator, PTTEP Australasia (Ashmore Cartier) Proprietary Limited (PTTEP AA), initially estimated the oil leakage to 400 barrels (or approximately 64 tonnes) of crude oil lost per day [3]. The uncontrolled release started on August 21, 2009, and crude oil continued to be released until November 3, 2009, when the well was brought under control. The Montara wellhead platform is located in Australian waters at the coordinate latitude $12^{\circ}40'20.5''$ N and longitude $124^{\circ}32'22.2''$ E. The depth of water was approximately 250 ft (76 m), as shown in Fig. 1. It is very close to Indonesian waters, only 310 km from the Timor island. This southernmost Indonesian archipelago has an important ecological and economical value. There are 3 355 352.82 hectare of marine protected areas in Sawu seas. Its coral ecosystems with 532 coral reef species and 11 species endemic with the total area of 633.39 km^2 [4] are a peculiar example. Following the Montara oil spill, the Indonesian Ministry of Marine and Fisheries Affairs initiated the development of ocean monitoring systems, in combination with the operational oceanography,

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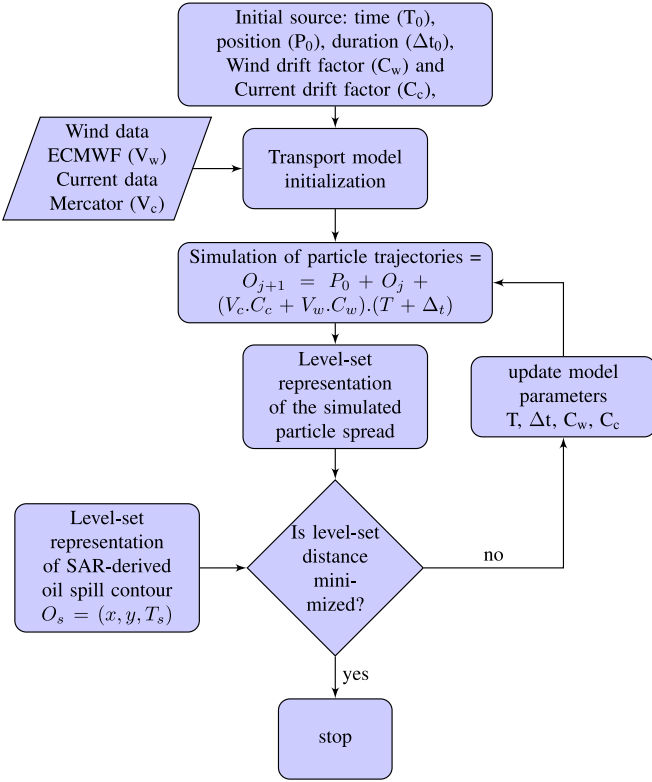


Fig. 2. Flowchart of the proposed approach for the assimilation of oil leakage parameters from satellite-based SAR observations.

especially within the framework of Infrastructure Development Space Oceanography (INDES0) project.

In this context, this study addresses the assimilation of satellite-based synthetic aperture radar (SAR) observations of oil spills into an oil trajectory model. It relies on Lagrangian-based simulations of the transport of particles at the sea surface and on the comparison of transport simulations and detected oil spills from implicit level-set-based representations. Using the Montara oil spill as a case study, we demonstrate the relevance of the proposed model and associated numerical scheme for the optimization of oil leakage parameters (namely the starting date of the leakage, its duration, the mean wind-related, and current-related drift coefficients). This paper is organized as follows. Section II describes the proposed approach. We report numerical experiments in Section III. Section IV further discusses the key features of the proposed model.

II. PROPOSED APPROACH

We report in Fig. 2 the synoptic sketch of the proposed approach for the assimilation of oil leakage parameters from satellite-based SAR observations. Our approach involves three main components.

- 1) A Lagrangian-based two-dimensional (2-D) transport model, which simulates the advection of particles at the sea surface under given wind and current conditions. This component is detailed in Section II-A.
- 2) The detection of oil spills from satellite-based SAR observations as detailed in Section II-B.

- 3) The definition of a similarity measure between simulated particle sets and oil spill detection, using implicit level-set representations, as detailed in Section II-C.

These three components are combined within an assimilation model detailed in Section II-D along with the associated numerical resolution.

A. Oil Spill Transport Model

The analysis of oil spill dynamics is an active research topic [5]. The considered models range from 2-D surface trajectory to 3D+t space-time dynamics and fate analysis. In general, an oil spill model combines a description of the oil spill, which may include the space-time extension of the oil spill as well as geochemical characteristics, and the undergoing weather conditions, typically wind and current conditions. In general, commonly used operational oil spill models include GNOME (General NOAA Operational Modeling Environment, <http://response.restoration.noaa.gov/>), MOTHY (French Operational Oil Spill Drift Forecast System, <http://www.meteorologie.eu.org/mothy/>), OSCAR (Oil Spill Contingency and Response, <http://www.sintef.no>), ADIOS2 (Automated Data Inquiry for Oil Spills, <http://response.restoration.noaa.gov>), OILMAP (Oil Spill Model and Response System, <http://www.aims.gov.au/pages/research/oil-map/oil-map01.html>), and OSIS (Oil Spill Identification System, <http://www.osis.biz/ss2.asp>) [6]. In Deepwater Horizon, as one of the biggest oil platform leakage accidents, the efforts of observation to detecting the existence of oil on the ocean surface and modeling of the oil spill trajectory are well documented [7]. One of the studies that was conducted for this accident is an assimilation of modeling and satellite-observed surface oil locations, where models were seeded with oil location inferred from satellite images with virtual particles and then these particles were advected with the surface velocity fields as forecast by numerical ocean circulation models [8]. The simulated surface oil trajectories were compared to the satellite observations in subsequent forecast cycles for veracity testing [9]. We consider some study applications in the case of the oil spill accident, i.e., two oil platforms in the Bohai Sea, China [10], and the Shell North Sea Gannet Alpha platform [11].

Here, our focus is on the assimilation of oil leakage parameters, and we consider a computationally tractable surface transport model. The oil spill leakage is characterized by the location of the source of the leakage, the type, and the quantity of oil. The considered transport model, namely the Mobidrift model [12] developed by Collecte Localisation Satellite (CLS), simulates the Lagrangian trajectories of particles at the sea surface for given wind and current conditions. In the Mobidrift model, fluids are modeled by a set of deterministic particles. For a source defined by a fixed oil platform, we provide to the Mobidrift model the location of the source, the starting time for the release of particles, and the duration of the leakage. For each particle released from the initial position, we apply the considered 2-D transport model to derive its trajectory up to the final date of the simulation. In our simulations, we typically consider about 500 particles released on a daily basis.

Formally, the Mobidrift model relies on the 2-D advection of particles by a surface displacement given by a weighted sum of tidal current and wind-induced current velocities

$$u_p^{\text{adv}} = C_c \cdot V_c + C_w \cdot V_w \quad (1)$$

where u_p^{adv} is the advection velocity, V_c is the sea surface current, and V_w is the sea surface wind speed at a height of 10 m. Coefficients C_c and C_w are the current and wind drift factors. In our study, as the sea surface wind and current data, we use the PSY4V2 MERCATOR system (1/12° horizontal resolution) with daily forecast production [13], and a 10-m wind field provided by the analysis of the European Centre for Medium-Range Weather Forecasts at a 0.140° spatial and 6-h temporal resolution [14].

The transport model is complemented with a diffusion component. Neglecting the viscous effects, the total velocity of a transported particle can be decomposed as

$$\frac{dx}{dt} = u_p^{\text{adv}}(x, t) + u_p^{\text{dif}}(x, t) \quad (2)$$

where $u_p^{\text{dif}}(x, t)$ is the diffusive component, which amounts to an isotropic Gaussian velocity, parameterized by standard deviation σ .

The simulation of the displacement of a particle between time t and time $t + \delta t$ resorts to the integration of (2) as follows:

$$x_p(t + \delta t) = x_p(t) + \int_t^{t+\delta t} u_p^{\text{adv}}(x_p, t) dt + \int_t^{t+\delta t} u_p^{\text{dif}}(x_p, t) dt. \quad (3)$$

It should be stressed that, through the stochastic diffusive component, the numerical integration results in a randomized simulation of trajectories. As a result, from multiple numerical integration, this transport model results in the simulation of the spread of an oil spill.

Wind drift factor C_w in (1) typically ranges from 0.01 to 0.06 [15]–[17]. In many applications of oil movement forecast, wind drift factor C_w is set to 0.03 [18], [19]. Regarding sea surface current, drift factor C_c typically ranges from 1.0 to 1.5 [16]. In this study, factors C_w and C_c are among the oil leakage parameters estimated to assimilate the satellite-based SAR observations and the simulated oil particles from the trajectory models.

B. SAR-Based Oil Spill Detection

Remote sensing can provide substantial support to routine surveillance in open ocean and coastal areas and has the advantage of being able to observe oil spill events in remote and often inaccessible areas. The high-resolution radar technology (the so-called synthetic aperture radar—hereafter SAR) on board satellites is widely used for operational oil spill surveillance [20], [21].

The information that is present in these data can be useful for extracting important indications for risk assessment, emergency management, and damage inventory [22]. Thanks to its very wide area coverage and its capability to work day/night independently of the cloud coverage, space-borne SAR technology

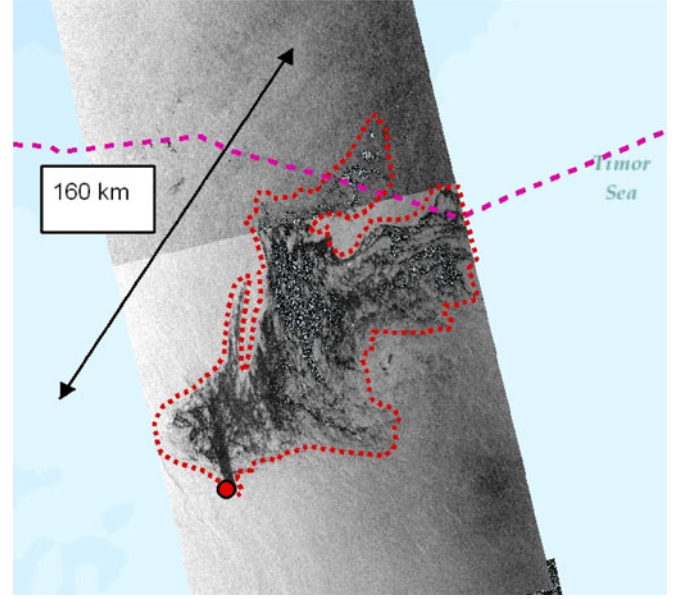


Fig. 3. SAR observation on September 2, 2009 by ENVISAT; red contour outlines the main patches of weathered oil on the sea surface, the red dot is the Montara wellhead position, and the purple outlines the border of Indonesia (above the line) and Australia water territory (below the line).

provides cost-efficient means. Flying at about 800-km altitude, SAR sensors emit electromagnetic (EM) waves with frequency ranging from 1 to 10 GHz. At these EM wavelengths, the sea surface backscatters the EM waves back to the sensor, depending on the sea state and radar acquisition geometry following Bragg theory. An oil spill dampens the sea surface roughness and basically induces a decrease of this EM backscattering compared to the surrounding clean water. Hence, the oil spill appears darker on the SAR image (see the red contour in Fig. 3). It is well established that reliable slick detections are performed for wind speeds in the approximate range of 2.5 to 12.5 m/s. In the case of low winds situation (below 2.5 m/s), the clean sea surface and the slick show similar radar backscattering and cannot be distinguished automatically. The operators in charge of analyzing the SAR images shall make the best use of any available contextual information such as ancillary data (sea surface temperature, wind field, chlorophyll-A concentration, etc.) in order to assess in the most efficient manner the situation at sea such as captured by the SAR sensor. A method was developed using μ as a logical scalar descriptor to map oil slicks under low to moderate wind conditions [23].

The oil properties at the sea surface are affected by weathering processes (evaporation, emulsification, photo-oxidation). The weathering of hydrocarbons is complex as it combines the diversity of petroleum products, which have very variable physical-chemical characteristics, with the variety of environmental conditions (wind and waves, sea temperature, etc.) As the SAR sensor is mostly sensitive to the sea surface roughness, slick with low viscosity or into the sea subsurface may not be detected.

In this study, the first SAR image covering the Montara event is used. It was observed on September 2, 2009, at 10:07 UTC by the European ENVISAT mission (see Fig. 3). The spatial

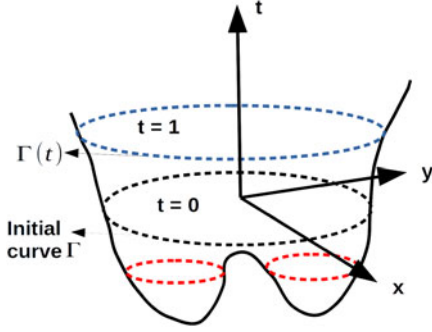


Fig. 4. Level-set [25] representation of closed contours in a 2-D plane: when the contours $t = 0$ (black, -) moving forward on $t = 1$ (blue, -) and if moving backward $t = -1$ (red, -).

resolution is $75 \text{ m} \times 75 \text{ m}$ with a swath width of 400 km. Since the blowout from the Montara wellhead platform first occurred on August 21, 2009, this image was taken 12 days after the accident. As seen by the SAR sensor, the oil slick is still connected to the wellhead platform, with estimated 160 km length.

It is worth noting that the resolution of SAR observation does not induce a dramatic evolution of the detection quality [24], nevertheless, spatial resolution of auxiliary data has a significant impact on the drift simulation and oil slick deformation. Hence, the quality of auxiliary data impacts the quality of oil leakage parameters estimation.

C. Similarity Measure Between Oil Spill Detection and Model Simulation

A key component of our approach is the definition of a measure of the goodness of the match between the detected oil spill and the simulation of transport model trajectories. This amounts to defining a similarity measure between a set of particles and a contour within a 2-D plane. We exploit implicit level-set representations of contours as introduced by Osher and Sethian [25]. As illustrated in Fig. 4, the level-set representation comes to defining a 2-D contour Γ as the zero-level-set of a function ϕ , that is to say the set of points of the plane at which the value of function ϕ is zero:

$$\Gamma = \{p \in \Omega \text{ such that } \phi(p) = 0\} \quad (4)$$

where Ω is the considered spatial region. As illustrated in Fig. 4, the level-set function ϕ is positive inside contour Γ and negative outside. Given function ϕ , the extraction of contour Γ simply comes to extracting the contour of a thresholded version of function ϕ at zero. Conversely, one can build the level-set representation of any contour in the plane as the solution of partial differential equation [25] using efficient numerical algorithms. A key interest of this level-set representation is its ability to represent nonconnected objects.

Level-set functions also provide a simple mean, without any topological constraint, to evaluate the similarity between any two contours Γ_1 and Γ_2 from a distance between their level-set representations [25]:

$$d(\Gamma_1, \Gamma_2) = \int_{\Omega} \|\phi_1(p) - \phi_2(p)\|^2 dp. \quad (5)$$

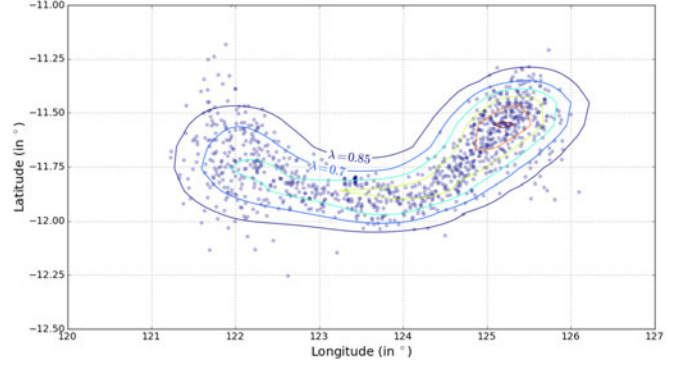


Fig. 5. Spatial density of a set of particles simulated from the Lagrangian transport model (2). The simulated particle set is depicted as a set of blue dots. We also display the contour of the kernel-based estimate of the spatial density of the simulated particle set.

This level-set representation directly applies to the contour of the satellite-based SAR detection of the oil spill. We also derive a level-set representation of the spread of the particle sets simulated by the considered Lagrangian transport model. Given a set of particles, we first derive the associated spatial density distribution $\mathcal{D}_S$ using a kernel-based estimation [26]. We then extract the contour Γ_S of the level-set of density distribution $\mathcal{D}_S$, which accounts for a predefined percentage of particles. Let us denote by λ this percentage and by $\mathcal{LL}_S(\nu)$ the level-set of $\mathcal{D}_S$ with respect to level ν :

$$\mathcal{LL}_S(\nu) = \{p \in \Omega \text{ such that } \mathcal{D}_S(p) > \nu\}. \quad (6)$$

Contour $\Gamma_S(\lambda)$ is then extracted as the contour of level-set $\mathcal{LL}_S(\nu^*)$ such that $|\mathcal{LL}_S(\nu^*)|/|\Omega| = \lambda$. We typically set λ to 0.85, such that the contour extracted to represent a particle set accounts for 85% of the simulated particles according to transport model defined by (3). As illustration, we show on Fig. 5, the contours of the kernel-based estimate created with different values for spatial density of simulated particles from the Lagrangian transport model.

Using the standard Euclidean distance between level-set functions, we define a distance between transport model simulations, characterized by density function $\mathcal{D}_S$ and associated reference contour $\Gamma_S(\lambda)$, and the detected oil spill Γ_{SAR} from the distance between their respective level-set functions $\phi_S(\lambda)$ and ϕ_{SAR} as follows:

$$d(\Gamma_S(\lambda), \Gamma_{\text{SAR}}) = \int_{\Omega} \|\phi_S(\lambda)(p) - \phi_{\text{SAR}}(p)\|^2 dp. \quad (7)$$

D. Estimation of Oil Leakage Parameters Assimilation of SAR Images

The general objective of our study is to develop a numerical approach for the estimation of oil leakage parameters given a satellite-based SAR detection of an oil spill and the location of the leakage source. Let us denote by θ the parameters to be estimated, namely leakage date d_L and duration Δ_L , and the mean surface current and wind drifts, respectively, C_e and C_w . Let us denote by Γ_{SAR} the contour of the detected oil spill from the SAR observation. We state the estimation of leakage

parameters θ as the maximization of the similarity between the detected oil spill and the simulated particle set from the Lagrangian transport model of (3) with respect to parameters θ . Numerically, it comes to solving for the following minimization:

$$\hat{\theta} = \arg \min_{\theta} d(\Gamma_S(\lambda|\theta), \Gamma_{\text{SAR}}) \quad (8)$$

where contour $\Gamma_S(\lambda|\theta)$ refers to the simulation of the Lagrangian transport model of (3) from the known leakage source with respect to parameter θ . The numerical resolution of this minimization exploits Powell's method [27] as we cannot derive the first-order derivatives of the considered function with respect to parameter θ . Powell's method iterates until convergence one-dimensional searches along conjugate directions, initialized as the basis vectors of the considered n -dimensional parameter space. Our implementation uses the scipy optimization package under Python environment.

III. APPLICATION TO MONTARA CASE STUDY

As a mean to investigate the sensitivity of transport model simulations with respect to oil leakage parameters, we first report assimilation results for 1-day oil leakage ranging from August 21 to September 2. This first simulation uses different dates of source of leakage with 1-day duration.

The result is given in Fig. 6(a), showing that the first 5 days of leakage give the minimum value of RMSE. For each date, we solve for minimization (8) to estimate wind and current drift factors C_w and C_C . We report in Figs. 7 and 8 the series of the optimal parameters C_w and C_C related to the simulations on Fig. 6(a). For the source of leakage August 21, 2009, with 1-day duration, the minimum RMSE was reached for the optimal drift factors $C_w = 0.493$ and $C_C = 1.221$. The values of optimal current drift factor range from 1.2 to 1.3 (the boundary given to Powell's method being $[0.1, 2]$). The retrieved optimal wind drift factors range from about 0.037 to 0.049 (the boundary for the optimization scheme being $[0.01, 0.06]$). Overall, these initial experiments point the greater contribution of the oil leakage, which occurred from the first 5 days, to the oil spill detected from the SAR observation on September 2, 2009.

Then we try to simulate the leakage with different duration. The result for simulation from a leakage start on August 21, 2009, with different duration is given in Fig. 6(b). This simulation shows that the rms error decreases as the duration increases. The minimum rms error was reached for the source of leakage August 21, 2009, with a duration of 12 days. Most of the surface of the detected oil spill is actually explained by the leakage occurring on the first 5 days, and happened for its duration (12 days for source of leakage August 21, 2009, 11 days for source of leakage August 22, 2009, 10 days for August 23, 2009, etc.) We will see the proof of this hypothesis later in Fig. 11.

As an illustration of Powell's optimization procedure, we illustrate in Fig. 9 the minimized cost function landscape and the points at which the cost function is actually evaluated for a 12-day leakage that started on August 21, 2009. The contribution of the wind is of minor importance compared to the current as the impact of C_w values is negligible.

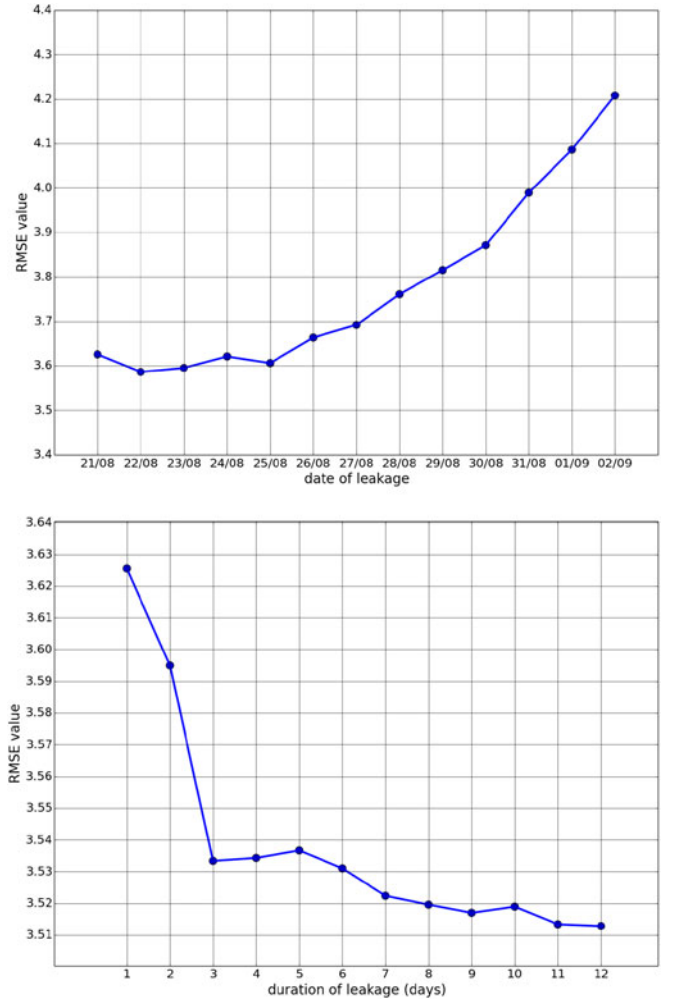


Fig. 6. Minimum RMSE of the level-set value of SAR images on September 2, 2009 with simulations of (a) 1-day oil leakage from August 21, 2009, to August 2, 2009, and (b) oil leakage started on August 21, 2009 with different duration from the Montara wellhead platform.

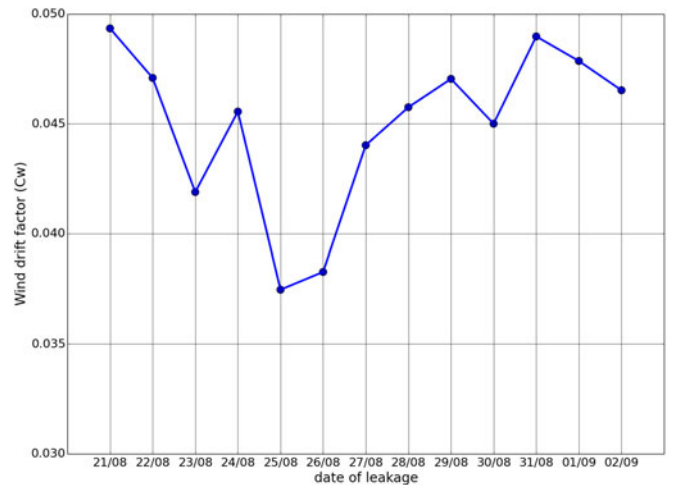


Fig. 7. Optimum value of wind drift factor C_w from simulations of 1-day oil leakage from the Montara wellhead platform from August 21, 2009, to September 2, 2009.

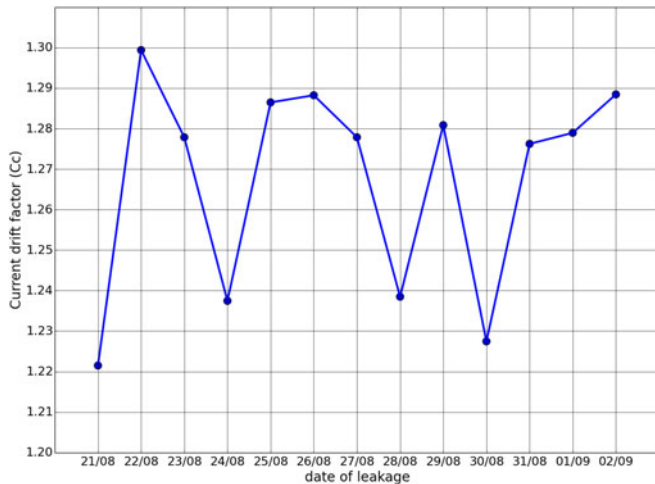


Fig. 8. Optimum value of current drift factor C_c from simulations of 1-day oil leakage from the Montara wellhead platform from August 21, 2009, to September 2, 2009.

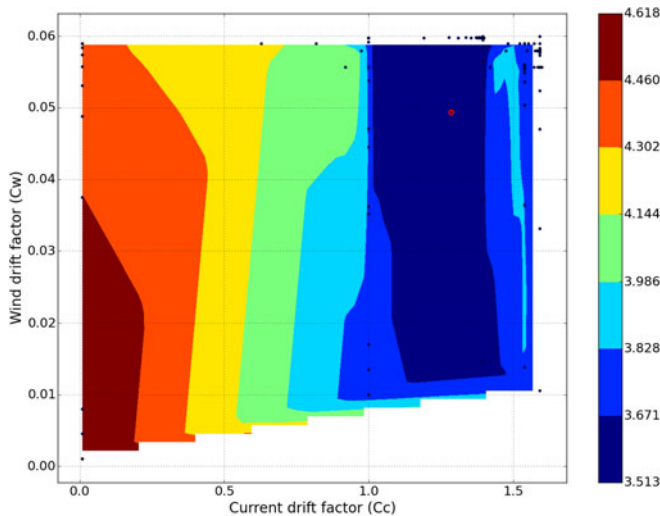


Fig. 9. Level-set-based distances between the SAR observation of the oil spill on September 2, 2009, and the Lagrangian-based drift simulations of 12-day oil leakage, which started on August 21, 2009, within Powell's optimization scheme for wind and current drift coefficients. Red dot shows the minimum RMSE value founded (3.513) with the value of $C_c = 1.285$ and $C_w = 0.493$.

We also report the level-set representations of the SAR-detected oil spill contour and of the transport model simulations for optimal drift factors $C_w = 0.493$ and $C_c = 1.221$ with 1-day duration in Fig. 10(a), as well as a comparison to the transport model simulations for optimal drift factors on simulations with 12 days duration $C_w = 0.493$ and $C_c = 1.285$ as shown in Fig. 10(b). Optimal drift factors with 12-day duration lead to a visually better match between model simulations and the SAR detection. Besides, such 1-day leakage hypothesis only provides a partial match to the observed oil spill spread.

We then carry out to the joint assimilation of both the oil leakage starting dates, its duration, and drift factors C_w and C_c . For each starting date and duration configuration, we proceed to minimization of (8). We report in Fig. 11 the resulting

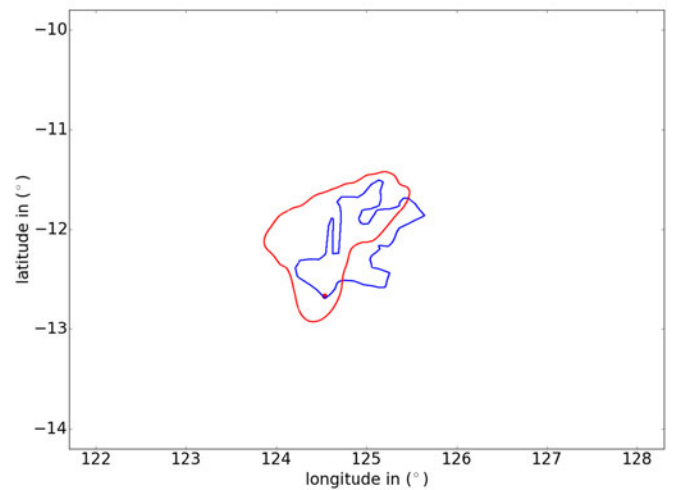
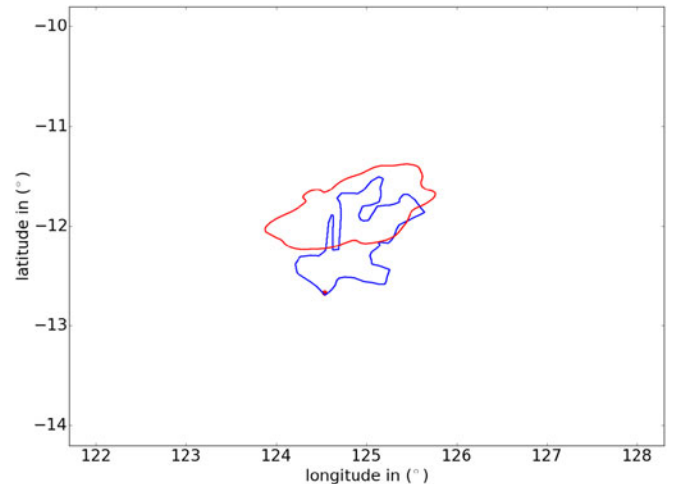


Fig. 10. Level-set-based representation of the SAR-derived detection of the oil spill on September 2, 2009, and the Lagrangian-based drift simulations for (a) a 1-day oil leakage on August 21, 2009, with optimal wind and current drift coefficients, $C_w = 49.3\%$ and $C_c = 122.1\%$ and (b) the Lagrangian-based drift simulations for a 12-day oil leakage, which started on August 21, 2009 with optimal wind and current drift coefficients, $C_w = 49.3\%$ and $C_c = 128.5\%$. The red dot shows the position of the Montara oil platform.

optimal level-set-based distances between the detected oil spill and transport model simulations as a function of the starting date of the oil leakage and its duration. The series of the level-set-based distance depict a clear increasing trend, which stresses the impact of the initial oil leakage on August 21. These results clearly support the existence of a continuous oil leakage for at least 5 days starting on August 21, 2009, with the minimum RMSE reached for oil leakage from oil leakage on August 21, 2009, with a duration of 12 days. Estimated optimal drift factors, respectively, $C_w = 0.493$ and $C_c = 1.285$, are realistic and in agreement with the experiments reported above and the previous work [15]–[17]. In Fig. 11, the comparison between transport model simulations and the SAR-derived oil spill detection on September 2, 2009, points out the improved match associated with the first 5 days with duration until the September 2 leakage hypothesis compared with the 1-day leakage hypothesis with nominal and optimized wind and current drift factors. It is

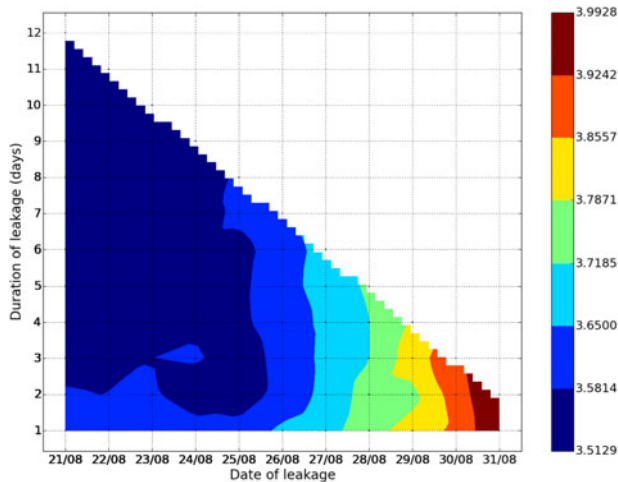


Fig. 11. Optimal level-set-based distances between the SAR observation of the oil spill on September 2, 2009 and the Lagrangian-based drift simulations of oil leakage with respect to leakage starting date (x-axis) and duration (y-axis). The optimal level-set-based distances are issued from Powell's optimization of wind and current drift coefficients.

worth noting that the Mobidrift simulation does not account for the concentration effect of the particles in the diffusion process so that it yields a similar level-set shape with the duration of leakage. That explains the flatness of the level-set distances of Fig. 11 with duration leakage between 2 and 12 days.

IV. CONCLUSION

We developed a novel framework for the assimilation of oil leakage parameters, including leakage starting data and duration along with wind and current drift factors, from an SAR-derived detection of an oil spill. This framework states the assimilation as the minimization of a level-set representation of the SAR-derived oil spill detection and of 2-D transport model simulations. We illustrated its relevance on the Montara oil spill case study. We showed that the observed oil spill on September 2, 2009, was mainly due to the oil leakage from August 21 to September 2 with majority of slick coming from the first 5 days of leakage (August 21–25, 2009). The estimated wind and current drift factors were in agreement with values previously reported in the literature. They were however significantly different from nominal parameters and shown to affect the match between the observed oil spill and transport model simulations.

This study opens new research avenues for the satellite-based operational monitoring of oil spills. Future work will further investigate the assimilation of oil leakage parameters from multirate observations as well as other oil leakage configurations, such as illegal oil discharges from vessels or natural oil seeps.

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